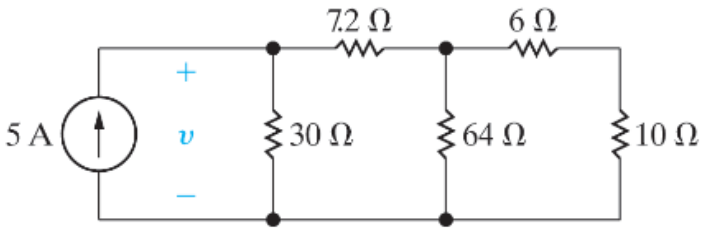
Chapter 3 Assessment Problems

3.1 For the circuit shown, find (a) the voltage v, (b) the power delivered to the circuit by the current source, and (c) the power dissipated in the $10\ \Omega$ resistor.



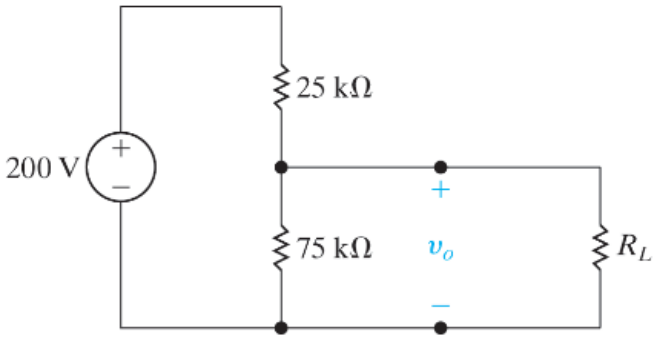
3.2

a. Find the no-load value of v_o in the circuit shown.

b. Find v_o when R_L is $150\ \text{k}\Omega$.

c. How much power is dissipated in the $25\ \text{k}\Omega$ resistor if the load terminals are accidentally short-circuited?

d. What is the maximum power dissipated in the $75\ \text{k}\Omega$ resistor?

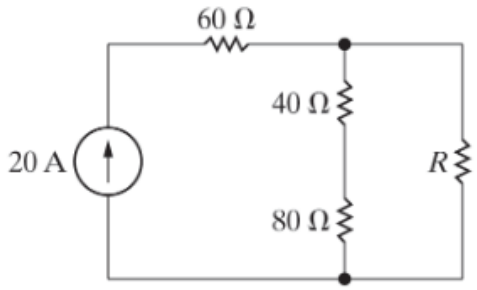


3.3

a. Find the value of R that will cause 4 A of current to flow through the 80 Ω resistor in the circuit shown.

b. How much power will the resistor R from part (a) need to dissipate?

c. How much power will the current source generate for the value of R from part (a)?

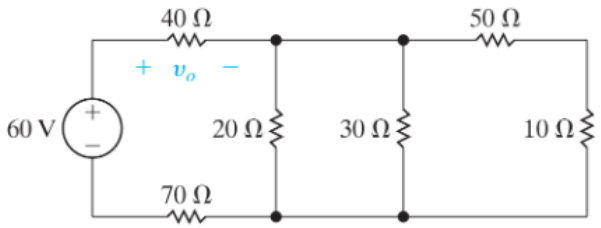


3.4

a. Use voltage division to determine the voltage v_o across the 40 Ω resistor in the circuit shown.

b. Use v_o from part (a) to determine the current through the 40 Ω resistor, and use this current and current division to calculate the current in the 30 Ω resistor.

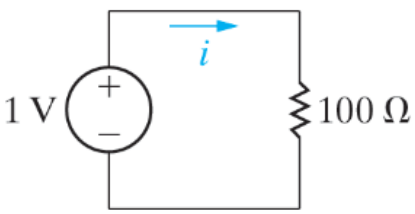
c. How much power is absorbed by the 50 Ω resistor?



3.5

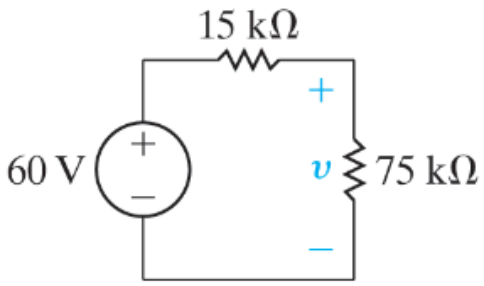
a. Find the current in the circuit shown.

b. If the ammeter in Example 3.8(a) is used to measure the current, what will it read?



3.6 **a.** Find the voltage v across the $75\ \text{k}\Omega$ resistor in the circuit shown.

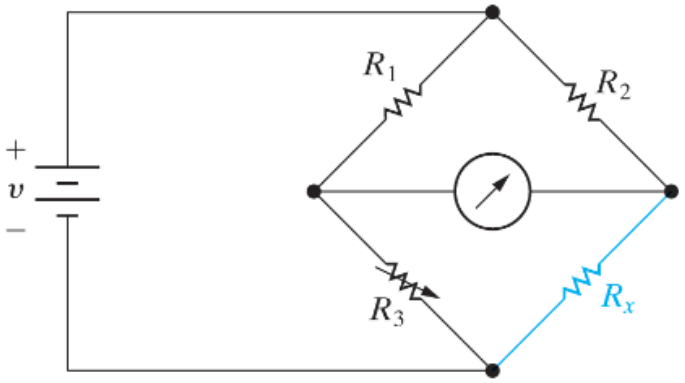
b. If the 150 V voltmeter of Example 3.9(a) is used to measure the voltage, what will be the reading?



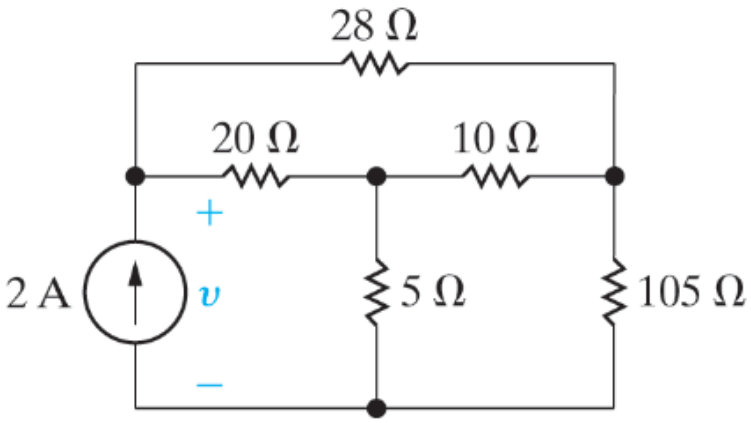
3.7 The bridge circuit shown is balanced when $R_1 = 100\ \Omega$, $R_2 = 1000\ \Omega$, and $R_3 = 150\ \Omega$. The bridge is energized from a 5 V dc source.

a. What is the value of R_x?

b. Suppose each bridge resistor is capable of dissipating 250 mW. Can the bridge be left in the balanced state without exceeding the power-dissipating capacity of the resistors, thereby damaging the bridge?



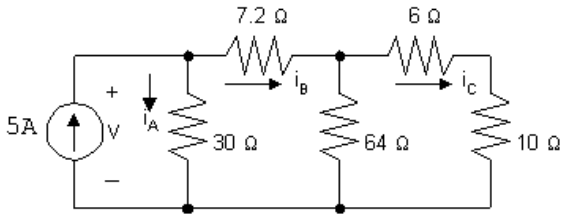
3.8 Use a Y-to-Δ transformation to find the voltage v in the circuit shown.



3 Simple Resistive Circuits

Assessment Problems

AP 3.1



Start from the right hand side of the circuit and make series and parallel combinations of the resistors until one equivalent resistor remains. Begin by combining the $6\,\Omega$ resistor and the $10\,\Omega$ resistor in series:

$$6\,\Omega + 10\,\Omega = 16\,\Omega.$$

Now combine this $16\,\Omega$ resistor in parallel with the $64\,\Omega$ resistor:

$$16\,\Omega \| 64\,\Omega = \frac{(16)(64)}{16+64} = \frac{1024}{80} = 12.8\,\Omega.$$

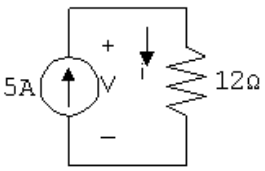
This equivalent $12.8\,\Omega$ resistor is in series with the $7.2\,\Omega$ resistor:

$$12.8\,\Omega + 7.2\,\Omega = 20\,\Omega.$$

Finally, this equivalent $20\,\Omega$ resistor is in parallel with the $30\,\Omega$ resistor:

$$20\,\Omega \| 30\,\Omega = \frac{(20)(30)}{20+30} = \frac{600}{50} = 12\,\Omega.$$

Thus, the simplified circuit is as shown:



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[a] With the simplified circuit we can use Ohm's law to find the voltage across both the current source and the $12\,\Omega$ equivalent resistor:

$v = (12\,\Omega)(5\text{ A}) = 60\text{ V}.$

[b] Now that we know the value of the voltage drop across the current source, we can use the formula $p = -vi$ to find the power associated with the source:

$p = -(60\text{ V})(5\text{ A}) = -300\text{ W}.$

Thus, the source delivers 300 W of power to the circuit.

[c] We now can return to the original circuit, shown in the first figure. In this circuit, $v = 60$ V, as calculated in part (a). This is also the voltage drop across the $30\,\Omega$ resistor, so we can use Ohm's law to calculate the current through this resistor:

$$i_A = \frac{60\text{ V}}{30\,\Omega} = 2\text{ A}.$$

Now write a KCL equation at the upper left node to find the current i_B:

$-5\text{ A} + i_A + i_B = 0 \qquad \text{so} \qquad i_B = 5\text{ A} - i_A = 5\text{ A} - 2\text{ A} = 3\text{ A}.$

Next, write a KVL equation around the outer loop of the circuit, using Ohm's law to express the voltage drop across the resistors in terms of the current through the resistors:

$-v + 7.2i_B + 6i_C + 10i_C = 0.$

So $\qquad 16i_C = v - 7.2i_B = 60\text{ V} - (7.2)(3) = 38.4\text{ V}.$

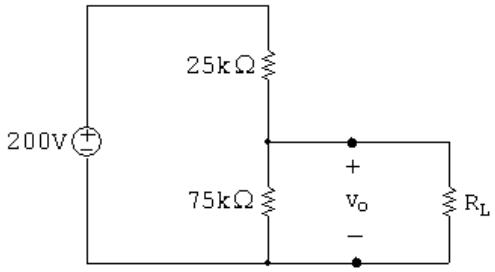
Thus $\qquad i_C = \dfrac{38.4}{16} = 2.4\text{ A}.$

Now that we have the current through the $10\,\Omega$ resistor we can use the formula $p = Ri^2$ to find the power:

$p_{10\,\Omega} = (10)(2.4)^2 = 57.6\text{ W}.$

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AP 3.2



[**a**] We can use voltage division to calculate the voltage v_o across the 75 kΩ resistor:

$$v_o(\text{no load}) = \frac{75{,}000}{75{,}000 + 25{,}000}(200\text{ V}) = 150\text{ V}.$$

[**b**] When we have a load resistance of 150 kΩ then the voltage v_o is across the parallel combination of the 75 kΩ resistor and the 150 kΩ resistor. First, calculate the equivalent resistance of the parallel combination:

$$75\text{ k}\Omega \| 150\text{ k}\Omega = \frac{(75{,}000)(150{,}000)}{75{,}000 + 150{,}000} = 50{,}000\,\Omega = 50\text{ k}\Omega.$$

Now use voltage division to find v_o across this equivalent resistance:

$$v_o = \frac{50{,}000}{50{,}000 + 25{,}000}(200\text{ V}) = 133.3\text{ V}.$$

[**c**] If the load terminals are short-circuited, the 75 kΩ resistor is effectively removed from the circuit, leaving only the voltage source and the 25 kΩ resistor. We can calculate the current in the resistor using Ohm's law:

$$i = \frac{200\text{ V}}{25\text{ k}\Omega} = 8\text{ mA}.$$

Now we can use the formula $p = Ri^2$ to find the power dissipated in the 25 kΩ resistor:

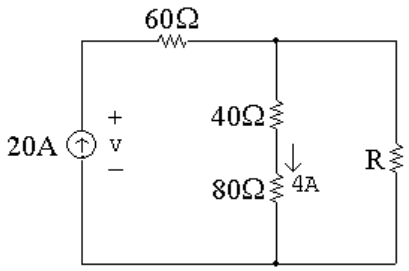
$$p_{25k} = (25{,}000)(0.008)^2 = 1.6\text{ W}.$$

[**d**] The power dissipated in the 75 kΩ resistor will be maximum at no load since v_o is maximum. In part (a) we determined that the no-load voltage is 150 V, so be can use the formula $p = v^2/R$ to calculate the power:

$$p_{75k}(\text{max}) = \frac{(150)^2}{75{,}000} = 0.3\text{ W}.$$

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AP 3.3



[a] We will write a current division equation for the current through the 80Ω resistor and use this equation to solve for R:

$$i_{80\Omega} = \frac{R}{R + 40\,\Omega + 80\,\Omega}(20\text{ A}) = 4\text{ A} \qquad \text{so} \qquad 20R = 4(R + 120).$$

$$\text{Thus} \qquad 16R = 480 \qquad \text{and} \qquad R = \frac{480}{16} = 30\,\Omega.$$

[b] With $R = 30\,\Omega$ we can calculate the current through R using current division, and then use this current to find the power dissipated by R, using the formula $p = Ri^2$:

$$i_R = \frac{40 + 80}{40 + 80 + 30}(20\text{ A}) = 16\text{ A} \qquad \text{so} \qquad p_R = (30)(16)^2 = 7680\text{ W}.$$

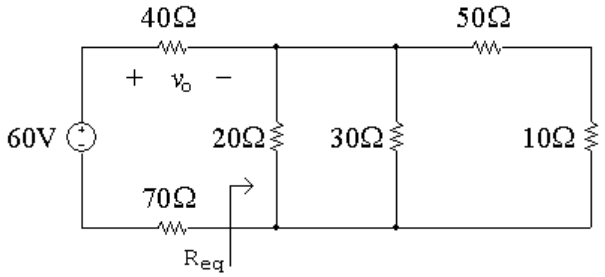
[c] Write a KVL equation around the outer loop to solve for the voltage v, and then use the formula $p = -vi$ to calculate the power delivered by the current source:

$$-v + (60\,\Omega)(20\text{ A}) + (30\,\Omega)(16\text{ A}) = 0 \qquad \text{so} \qquad v = 1200 + 480 = 1680\text{ V}.$$

$$\text{Thus,} \qquad p_{\text{source}} = -(1680\text{ V})(20\text{ A}) = -33{,}600\text{ W}.$$

Thus, the current source generates 33,600 W of power.

AP 3.4



[a] First we need to determine the equivalent resistance to the right of the $40\,\Omega$ and $70\,\Omega$ resistors:

$$R_{\text{eq}} = 20\,\Omega \| 30\,\Omega \| (50\,\Omega + 10\,\Omega) \qquad \text{so} \qquad \frac{1}{R_{\text{eq}}} = \frac{1}{20\,\Omega} + \frac{1}{30\,\Omega} + \frac{1}{60\,\Omega} = \frac{1}{10\,\Omega}.$$

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Thus, $R_{\text{eq}} = 10\,\Omega$.

Now we can use voltage division to find the voltage v_o:

$$v_o = \frac{40}{40 + 10 + 70}(60\text{ V}) = 20\text{ V}.$$

[b] The current through the $40\,\Omega$ resistor can be found using Ohm's law:

$$i_{40\Omega} = \frac{v_o}{40} = \frac{20\text{ V}}{40\,\Omega} = 0.5\text{ A}.$$

This current flows from left to right through the $40\,\Omega$ resistor. To use current division, we need to find the equivalent resistance of the two parallel branches containing the $20\,\Omega$ resistor and the $50\,\Omega$ and $10\,\Omega$ resistors:

$$20\,\Omega \| (50\,\Omega + 10\,\Omega) = \frac{(20)(60)}{20 + 60} = 15\,\Omega.$$

Now we use current division to find the current in the $30\,\Omega$ branch:

$$i_{30\Omega} = \frac{15}{15 + 30}(0.5\text{ A}) = 0.16667\text{ A} = 166.67\text{ mA}.$$

[c] We can find the power dissipated by the $50\,\Omega$ resistor if we can find the current in this resistor. We can use current division to find this current from the current in the $40\,\Omega$ resistor, but first we need to calculate the equivalent resistance of the $20\,\Omega$ branch and the $30\,\Omega$ branch:

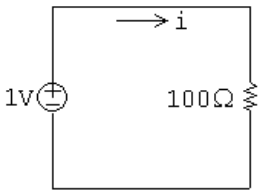
$$20\,\Omega \| 30\,\Omega = \frac{(20)(30)}{20 + 30} = 12\,\Omega.$$

Current division gives:

$$i_{50\Omega} = \frac{12}{12 + 50 + 10}(0.5\text{ A}) = 0.08333\text{ A}.$$

Thus, $p_{50\Omega} = (50)(0.08333)^2 = 0.34722\text{ W} = 347.22\text{ mW}.$

AP 3.5 **[a]**

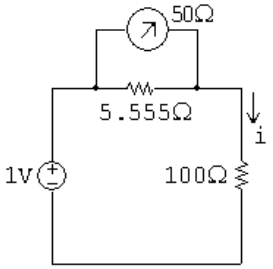


We can find the current i using Ohm's law:

$$i = \frac{1\text{ V}}{100\,\Omega} = 0.01\text{ A} = 10\text{ mA}.$$

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[b]

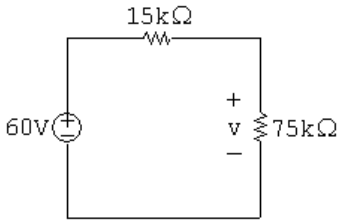


$R_m = 50\,\Omega \| 5.555\,\Omega = 5\,\Omega.$

We can use the meter resistance to find the current using Ohm's law:

$$i_{\text{meas}} = \frac{1\text{ V}}{100\,\Omega + 5\,\Omega} = 0.009524 = 9.524\text{ mA}.$$

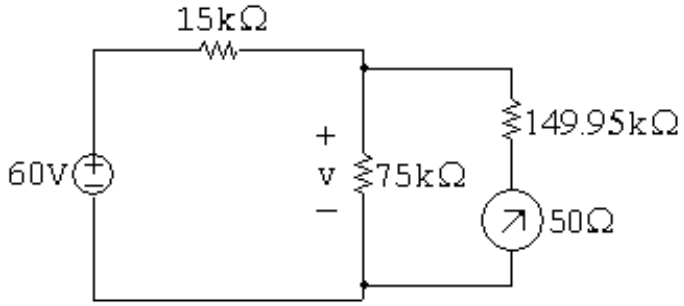
AP 3.6 **[a]**



Use voltage division to find the voltage v:

$$v = \frac{75{,}000}{75{,}000 + 15{,}000}(60\text{ V}) = 50\text{ V}.$$

[b]



The meter resistance is a series combination of resistances:

$R_m = 149{,}950 + 50 = 150{,}000\,\Omega.$

We can use voltage division to find v, but first we must calculate the equivalent resistance of the parallel combination of the 75 kΩ resistor and the voltmeter:

$$75{,}000\,\Omega \| 150{,}000\,\Omega = \frac{(75{,}000)(150{,}000)}{75{,}000 + 150{,}000} = 50\text{ k}\Omega.$$

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Thus, $\qquad v_{\text{meas}} = \dfrac{50{,}000}{50{,}000 + 15{,}000}(60\text{ V}) = 46.15\text{ V}.$

AP 3.7 **[a]** Using the condition for a balanced bridge, the products of the opposite resistors must be equal. Therefore,

$$100R_x = (1000)(150) \qquad \text{so} \qquad R_x = \frac{(1000)(150)}{100} = 1500\,\Omega = 1.5\text{ k}\Omega.$$

[b] When the bridge is balanced, there is no current flowing through the meter, so the meter acts like an open circuit. This places the following branches in parallel: The branch with the voltage source, the branch with the series combination R_1 and R_3 and the branch with the series combination of R_2 and R_x. We can find the current in the latter two branches using Ohm's law:

$$i_{R_1,R_3} = \frac{5\text{ V}}{100\,\Omega + 150\,\Omega} = 20\text{ mA}; \qquad i_{R_2,R_x} = \frac{5\text{ V}}{1000 + 1500} = 2\text{ mA}.$$

We can calculate the power dissipated by each resistor using the formula $p = Ri^2$:

$$p_{100\Omega} = (100\,\Omega)(0.02\text{ A})^2 = 40\text{ mW};$$

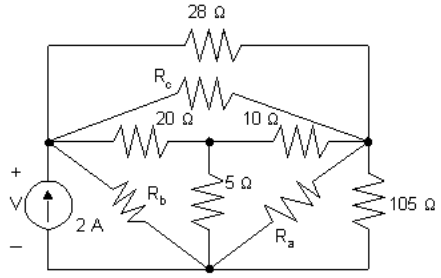
$$p_{150\Omega} = (150\,\Omega)(0.02\text{ A})^2 = 60\text{ mW};$$

$$p_{1000\Omega} = (1000\,\Omega)(0.002\text{ A})^2 = 4\text{ mW};$$

$$p_{1500\Omega} = (1500\,\Omega)(0.002\text{ A})^2 = 6\text{ mW}.$$

Since none of the power dissipation values exceeds 250 mW, the bridge can be left in the balanced state without exceeding the power-dissipating capacity of the resistors.

AP 3.8 Convert the three Y-connected resistors, $20\,\Omega$, $10\,\Omega$, and $5\,\Omega$ to three Δ-connected resistors $R_{\text{a}}, R_{\text{b}}$, and R_{c}. To assist you the figure below has both the Y-connected resistors and the Δ-connected resistors



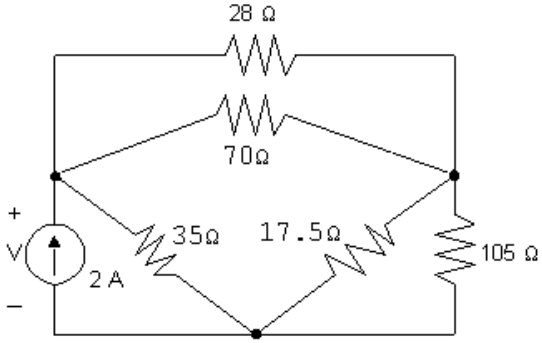
$$R_{\text{a}} = \frac{(5)(10) + (5)(20) + (10)(20)}{20} = 17.5\,\Omega;$$

$$R_{\text{b}} = \frac{(5)(10) + (5)(20) + (10)(20)}{10} = 35\,\Omega;$$

$$R_{\text{c}} = \frac{(5)(10) + (5)(20) + (10)(20)}{5} = 70\,\Omega.$$

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The circuit with these new Δ-connected resistors is shown below:



From this circuit we see that the $70\,\Omega$ resistor is parallel to the $28\,\Omega$ resistor:

$$70\,\Omega \| 28\,\Omega = \frac{(70)(28)}{70+28} = 20\,\Omega.$$

Also, the $17.5\,\Omega$ resistor is parallel to the $105\,\Omega$ resistor:

$$17.5\,\Omega \| 105\,\Omega = \frac{(17.5)(105)}{17.5+105} = 15\,\Omega.$$

Once the parallel combinations are made, we can see that the equivalent $20\,\Omega$ resistor is in series with the equivalent $15\,\Omega$ resistor, giving an equivalent resistance of $20\,\Omega + 15\,\Omega = 35\,\Omega$. Finally, this equivalent $35\,\Omega$ resistor is in parallel with the other $35\,\Omega$ resistor:

$$35\,\Omega \| 35\,\Omega = \frac{(35)(35)}{35+35} = 17.5\,\Omega.$$

Thus, the resistance seen by the 2 A source is $17.5\,\Omega$, and the voltage can be calculated using Ohm's law:

$$v = (17.5\,\Omega)(2\text{ A}) = 35\text{ V}.$$

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